

Cosmology Research Programme:

Investigations, Results and Pending Tests

A five-page technical summary of the PLAMB/FR, supernova, SPARC, SU(2), Quaia, HHT, SkyMap and BBN work

Prepared for the PLAMB research programme

19 July 2026 - current evidence summary

Research status

The programme now has reproducible tests across standard candles, galaxy rotation curves, sky anisotropy and primordial abundances.

The 19 July extension adds an outcome-aware raw-MU near-tie, a covariant clock-distance-ruler relation, a failed physical BAO closure, and frozen DESI/eBOSS Lyman-alpha validation. No stream currently passes every promotion gate.

The governing boundaries are: SPARC subset wins, not a full-sample win; rotation curves do not identify antimatter; Quaia is a follow-up candidate; $\gamma_c = 2.3$ is post hoc; the radial/transverse powers are an empirical Lambda-CDM-like coordinate; and the DESI DR2 Lyman-alpha stress is cross-led but not independently replicated at 1%.

Abstract

The research began with phenomenological tests of whether redshift-distance relations, light propagation or clock rate effects could imitate part of the standard cosmological signal.

It expanded into a SPARC optical-depth representation of the radial acceleration relation, SU2/SU2R background fits, Quaia angular scans, SkyMap and Hilbert-Huang Transform (HHT) residual searches, and finally Big Bang nucleosynthesis (BBN) calculations using LINX.

The strongest contributions are methodological: full-covariance and hierarchical likelihoods, preregistered frames and nulls, non-centred sampling, charge-identifiability and covariant-closure derivations, injection recovery, frozen hold-outs, immutable manifests and independent numerical validation.

Positive local patterns exist, but calibration, spatial clustering, predictive stress, sampler divergences, failed physical closure and primordial-abundance incompatibilities prevent discovery language.

The next phase should emphasise independent catalogues, action-level clock/photon/ruler physics, native high-redshift likelihoods, matched nuisance structures and first-principles derivations.

Keywords: PLAMB; FR cosmology; supernovae; SPARC; SU(2); Quaia; HHT; BBN; LINX; reproducibility.

1. Supernovae, Clock-Distance Closure and BAO

Scope. Pantheon+SH0ES, DES-SN5YR/Dovekie and Union3.1 supply released distance moduli and covariance; DESI DR2/DR1 and eBOSS DR16 supply transverse and radial BAO controls.

Principal programmes: `run_joint_rawmu_hierarchical_offset_likelihood.py` and the July 19 gamma, covariant-closure, missing-degree, radial/transverse and Lyman-alpha audits.

The default supernova diagnostic remains the released-covariance hierarchical raw-MU likelihood with zHD primary. The fixed $\gamma_c = 2.3$ branch is explicitly outcome-aware.

Frozen July 19 path and BAO definitions

$$\begin{aligned} k(z) &= (1 + 2.3 z)/(1 + z)^p \\ \chi(z) &= \int_0^z k(u) du \\ D_M(z) &= q \chi(z)/(1 + z)^{b_{\text{perp}}} \\ D_H(z) &= q k(z)/(1 + z)^{b_{\text{parallel}}} \end{aligned}$$

Here p is fixed by the raw-MU path, q is the common BAO scale, and b_{perp} and b_{parallel} are empirical transverse and radial powers.

Table 1. Supernova, closure and high-redshift tests completed through 19 July 2026

Completed test	Principal readout	Standing interpretation
Released-total raw-MU	$p = 0.797429$; Delta BIC = +0.011958	PATH and flat Lambda-CDM are tied; γ_c is post hoc
Calibration-budget sensitivity	Delta BIC range -0.816882 to +0.565488	Stable near-tie within external budgets
Preregistered frames	Pantheon+ and DES: zHD primary; zHEL/zCMB controls	zHD remains the default readout
Covariant closure	Best tested one-degree closure chi-squared = 2684.670; Lambda-CDM = 10.271	Kinematic relation derived; physical adapter fails
DR2 Lyman-alpha	Combined $p = 0.001705$; cross $p = 0.002975$; auto $p = 0.065260$	Cross-led frozen-prediction failure
External/method controls	Multipoles $p = 0.005170$; DR1 $p = 0.092834$; eBOSS HPD = 0.079816	Same-spectra robustness; no 1% independent replication

1.1 Current conclusion

The released-total raw-MU PATH and flat Lambda-CDM fits are tied. External-budget reconstructions preserve that near-tie, but $\gamma_c = 2.3$ was selected after inspecting the outcomes and the high-redshift predictive score is slightly worse.

A covariant relation now connects T , D_L , D_M , D_H and ruler evolution, but PLAMB does not derive the required atomic-frequency $Q(T)$, photon/luminosity $\Xi(T)$ or ruler $B(T)$ laws. Isotropic ruler evolution cannot repair the radial/transverse ratio.

Frozen lower-redshift PLAMB is rejected by the compressed DESI DR2 Lyman-alpha posterior, led by the forest-quasar cross-correlation. The failure survives a same-spectra multipole control but not a 1% independent-survey gate in DESI DR1 or native eBOSS DR16. zHD remains primary.

2. Galaxy Acceleration and Angular Structure

2.1 SPARC optical-depth branch

Optical-depth representation of the radial acceleration relation

$$\begin{aligned}\tau &= [g_{\text{bar}}/(\kappa c H_0/(2\pi))]^p \\ \nu &= [1 - \exp(-\tau)]^{(-1)} \\ g_{\text{obs}} &= \nu g_{\text{bar}}\end{aligned}$$

The filtered all-Q2 family contains 157 galaxies and 3,001 rotation points after removing six preregistered persistent-negative systems.

The four-chain non-centred NUTS run achieved maximum rank $R\text{-hat}$ 1.00259 for RAR and 1.00283 for PLAMB, but failed the locked zero-divergence criterion with 3 and 6 divergences respectively.

The fitted PLAMB exponent remains near one half ($p = 0.51892 \pm 0.00494$), but the all-Q2, low-acceleration outer-point and inclination predictive gates remain failures. The standing conclusion is subset wins, not a full-sample win.

Complete FR charge conjugation swaps the matter and antimatter background contributions while preserving total background magnitude and $|\chi|$. The implemented rotation-curve likelihood is therefore charge-even and cannot classify an isolated SPARC galaxy as matter or antimatter.

SPARC/FR gate

The preregistered 2MRS/NED environment test found a suggestive negative partial association in 66 galaxies ($\rho = -0.3073$, $p = 0.01180$), but predictive improvement was only 2.40%; the locked development gate failed and all five replication galaxies stayed sealed.

An independently published Kourkchi-Tully group-catalogue predictor was null in 65 galaxies ($\rho = 0.04768$, $p = 0.7044$) and worsened held-out RMSE by 5.41%.

Injection-recovery gives only 36.50% empirically recalibrated joint power at $|\rho| = 0.307254$; approximately $|\rho| = 0.5$ is needed for at least 80% recovery.

2.2 Quaia/SU2 angular tests

Controlled angular residual model: dipole plus survey templates

$$r_i = a + d \cdot n_i + \sum_k \gamma_k T_{ik} + \epsilon_i$$

A redshift-window candidate near $z = 1.0\text{--}1.5$ survived several object-level checks, but the full all-window look elsewhere probability was ordinary (p_{global} about 0.43 for the original strongest row).

After catalogue-quality matching the targeted family had $p = 1$.

Spatial-block mocks gave family $p = 0.0135$, 0.0923 and 0.2326 for 8, 12 and 16 degree cells, so the mode fails the requirement to remain below 1% across cluster scales.

The locked direction has template $R^2 = 0.1636$ and variance-inflation factor 1.196, with strongest correlation -0.4044 to a Gaia scan-angle template.

Injection-recovery exposed an important refinement: at the observed 1x amplitude, strict physical recovery after scan-law and colour control was 0.2833, while the final BIC gate accepted only 0.0011; 0.2823 of trials were physically recovered and rejected only by BIC.

The current template gate is therefore under-powered and collinear as well as non-confirmatory. The correct status is targeted follow-up candidate only, with no SU2 detection or headline anisotropy claim.

2.3 SkyMap and HHT archive

The historical SkyMap/Quaia and CMB-supernova HHT work has been curated with code, extracted reports and representative figures.

HHT identified intrinsic-mode-function structure, but shuffled-order controls were ordinary: extended SU2-minus-Lambda-CDM maximum and summed-energy upper-tail probabilities were 0.9703 and 0.9604.

Archive this as method development and candidate-feature evidence; a new preregistered probe separated analysis is required.

3. Primordial-Abundance Tests

3.1 FR/LINX validation

The FR investigation was translated into a standard expanding-background LINX clock/radiation proxy and scanned over 30,503 immutable key-network states.

A preregistered 92-evaluation validation matrix then tested $\text{rtol} = 10^{-7}$, $\text{atol} = 10^{-11}$, 300/600 interpolation samples, key/small/full PRIMAT 2023 networks, updated neutron-lifetime states, the three main deuterium-sensitive rates and five omitted Li/Be channels.

All 92 evaluations completed with no missing or failed rows.

Numerical and local deuterium-theory readout

$$\begin{aligned}\max |\Delta(10^5 \text{ D/H})| &= 5.51 \times 10^{-4} \\ \max |\Delta Y_p| &= 6.53 \times 10^{-5} \\ \max |\Delta(10^{10} \text{ Li/H})| &= 1.71 \times 10^{-3} \\ \sigma_{\text{D,theory}} &= 0.02608\end{aligned}$$

at the strict full-network baseline

Table 2. Publication-grade FR/LINX validation rows

Strict full-network state	10^5 D/H	Y_p	10^{10} Li/H	Registered χ^2
Baseline	2.434914	0.2469325	5.637694	295.6 (reference scale)
Best registered direct-Li	2.585689	0.248639	4.647982	210.57

There are zero registered and zero updated-current simultaneous D+He+Li two-standard-deviation passes.

Full minus-key corrections are at most 0.000649 in D, 1.48×10^{-5} in helium and 0.0580 in lithium.

The largest omitted channel lithium response, Be7daap, is 0.06079 at baseline, far below the residual excess.

The main statistical correction is instead the D theory budget: 0.02608 is about 90% of the registered observational error and must be marginalised jointly with rate pulls.

3.2 Literature-defined SU(2)-CMB bridge

High-temperature SU(2)-CMB benchmark

$$\begin{aligned}n_{\text{u,CMB}} &= 4^{(-1/3)} &= 0.629960525 \\ H_{\text{SU2CMB}}/H_{\text{SBBN}} &= 1.25326 \text{ at } T_{\text{gamma}} = 1 \text{ MeV} \\ N_{\text{eff,SU2CMB}} &= 20.373 \text{ (LINX diagnostic)}\end{aligned}$$

The separately registered 62-point bridge propagates the published high-temperature SU(2)-CMB equation of state through LINX while retaining standard weak and nuclear microphysics.

At fixed reference baryon density it predicts $\text{D/H} = 0.64441 \times 10^{-5}$, $Y_p = 0.3165$ and $\text{Li/H} = 26.476 \times 10^{-10}$.

The best D+He row at eta factor 0.48 still has $\chi^2_{\text{D+He}} = 501.41$; the best row inside the published baryon prior has total $\chi^2 = 3602.7$.

No abundance pass occurs.

This is not the repository's late-time SU2/SU2R fluid and supplies no first-principles gauge-matter changes to masses, binding energies, Q-values or rates.

4. Pending Tests and Research Path

The next programme should be narrower, preregistered and designed around independent falsification.

The following order gives the greatest gain in research standing per unit computation.

1. Quaia cross-catalogue validation

Repeat the $z = 1.0-1.5$ search in an independent quasar catalogue with the mask, windows, latitude cuts and spatial-block null fixed in advance; require global significance across block scales, stable amplitude/direction and coherent $l = 1-3$ structure.

2. Injection-calibrated angular likelihood

Replace the fixed Delta-BIC gate with a power-calibrated rule that jointly models dust, depth, Gaia scanning geometry, WISE colour and catalogue quality; require held-out SU2 scalar prediction to beat template-only prediction.

3. Action-level clock-distance-ruler theory

Derive an action or field equation for T and specify the matter/photon metrics, $Q(T)$, $\chi(T)$, the drag-epoch sound horizon and $B(T)$ before fitting another distance dataset; one parameter set must recover redshift, D_L , D_M and D_H .

4. Native high-redshift BAO validation

Obtain or reconstruct the native DESI DR2 correlation likelihood. Preregister auto/cross, quasar-redshift, proximity and continuum nuisance tests; require independent high-redshift replication before interpreting a residual as physical.

5. SPARC prediction, identifiability and power gate

Retune both non-centred models to zero divergences, run grouped galaxy-held-out prediction, prohibit matter/antimatter labels without a signed external field, and keep the five reserved galaxies sealed until an independent design has at least 80% calibrated joint power.

6. BBN hierarchy and first-principles models

Fit baryon density, neutron lifetime, rate pulls and stellar lithium depletion with matched likelihoods. FR must derive $T(t)$, $\rho(T)$, $c(T)$, masses, Q -values, binding energies and rates; keep the SU(2)-CMB bridge separate without an action-level link.

7. HHT/SkyMap confirmatory design

Preregister probe-separated residuals, bands and masks; use phase-randomised and order-preserving surrogates, global correction and held-out redshift ranges. Treat the existing archive as a hypothesis generator.

4.1 Reproducibility and evidence

Code, small results, reports, locks and manifests are curated at github.com/cliveboyd/Cosmology under `github_export/`. Large inputs remain checksum-referenced; future papers should cite commits, preregistrations, programme versions, releases and claim gates.

Evidence index: [1] raw-MU hierarchy/frame audit (14 Jul); [2] SPARC NUTS (16 Jul); [3] Quaia spatial-block audit (16 Jul); [4] FR/LINX matrix; [5] SU(2)-CMB bridge (17 Jul); [6] FR charge/environment/power (18 Jul); [7] γ /closure/missing-degree/radial-transverse/Lyman-alpha sequence (19 Jul). Full paths: `github_export/results/README.md`.

Conclusion

The programme now has a defensible map of survival and failure.

Highest-value work is an action-level clock/photon/ruler theory, native and independent high-redshift BAO validation, independent Quaia validation, zero-divergence held-out SPARC prediction, a powered independent environment test, and first-principles BBN models.